2025 SSMO Accuracy Round Solutions

SMO Team

AR 1: An infinite sequence of real numbers $a_1, a_2, \ldots$ satisfies

$$a_n = a_1 + a_2 + \cdots + a_{n-1}$$

for all positive integers $n > 1$. Given that $a_{20} = 25$, find a_{25}.

Answer: $\boxed{800}$

Solution: Note that for every positive integer $n > 2$, by the given recurrence relation, we have

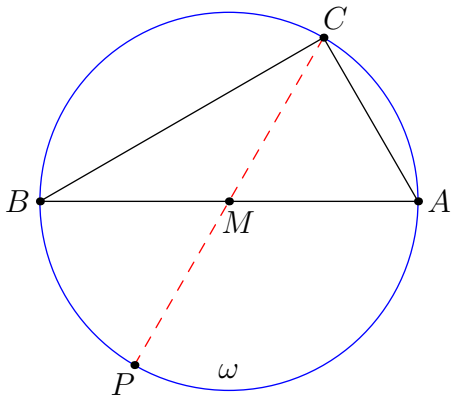
$$\begin{aligned} a_n &= a_{n-1} + a_{n-2} + \cdots + a_1 \\ &= (a_{n-2} + \cdots + a_1) + a_{n-2} + \cdots + a_1 \\ &= 2(a_{n-2} + \cdots + a_1) \\ &= 2a_{n-1}. \end{aligned}$$

Therefore, $a_{25} = 2a_{24} = 4a_{23} = 8a_{22} = 16a_{21} = 32a_{20} = \boxed{800}$.

AR 2: Let ABC be a triangle with circumcircle ω. The midpoint of AB is M, and the line CM intersects ω again at P. Given $\angle BMC = 120^\circ$, $\triangle BMC$ is isosceles, and $BC = 20$, the length of PM can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{26}$

Solution:



To begin, note that $\angle AMC = 60^\circ$ and $AM = BM = MC$. This means that $\triangle AMC$ is equilateral. Now, observe that $\angle BCM = \frac{1}{2}(180^\circ - 120^\circ) = 30^\circ$. Thus, $\angle ACB = 90^\circ$. Because $\triangle AMC$ is equilateral, $\angle BAC = 60^\circ$, implying that $\triangle ABC$ is a 30-60-90 triangle.

The circumcenter of a right triangle is the midpoint of its hypotenuse, so M is the center of ω and PM is a radius of ω. Therefore, $PM = AM$, since AM is also a radius of ω. By 30-60-90 triangle properties, $AB = \frac{2}{\sqrt{3}} \cdot BC = \frac{40\sqrt{3}}{3}$, so

$$PM = AM = \frac{1}{2} \cdot AB = \frac{20\sqrt{3}}{3} \implies 20 + 3 + 3 = \boxed{26}.$$

AR 3: Nonnegative real numbers x, y, and z satisfy

$$\frac{\sqrt{x} + 13}{y} = \frac{\sqrt{y} + 29}{z} = \frac{\sqrt{z} + 46}{x} = 2$$

and

$$\frac{\sqrt{x} + \sqrt{y} + \sqrt{z}}{x + y + z} = \frac{6}{25}.$$

Find the value of $x + y + z$.

Answer: $\boxed{50}$

Solution: From the first chain of equalities in the problem statement, when we clear any appearing fractions, we have

$$\begin{aligned} \sqrt{x} &= 2y - 13 \\ \sqrt{y} &= 2z - 29 \\ \sqrt{z} &= 2x - 46. \end{aligned}$$

Adding these three equations together, we obtain

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 2(x + y + z) - 88.$$

Plugging this into the last equation in the problem statement, we have

$$\frac{2(x + y + z) - 88}{x + y + z} = \frac{6}{25}.$$

Solving, we have $x + y + z = \boxed{50}$.

AR 4: Let a, b, and c be the roots of the polynomial $x^3 + 4x^2 - 3x - 4$. The value of

$$\left| \frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b} \right|$$

can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{21}$

Solution: Let $f(x) = x^3 + 4x^2 - 3x - 4 = (x - a)(x - b)(x - c)$. From Vieta's formulas, $a + b + c = -4$, so $b + c = -4 - a$, $c + a = -4 - b$, and $a + b = -4 - c$. So,

$$\begin{aligned} \frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b} &= \frac{a(c + a)(a + b) + b(a + b)(b + c) + c(b + c)(c + a)}{(a + b)(b + c)(c + a)} \\ &= \frac{a(-4 - b)(-4 - c) + b(-4 - c)(-4 - a) + c(-4 - a)(-4 - b)}{(-4 - a)(-4 - b)(-4 - c)} \\ &= \frac{16(a + b + c) + 8(ab + bc + ca) + 3abc}{f(-4)}. \end{aligned}$$

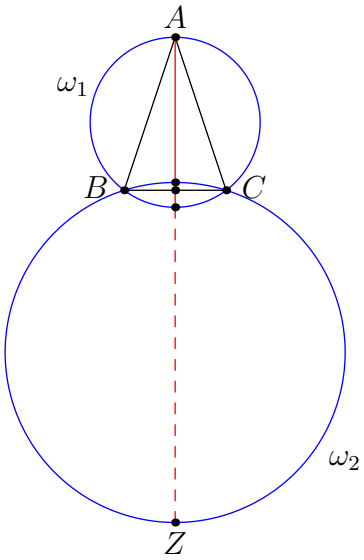
We have $f(-4) = 8$, and from Vieta's formulas, $a+b+c = -4$, $ab+bc+ca = -3$, and $abc = 4$. Substituting, we have

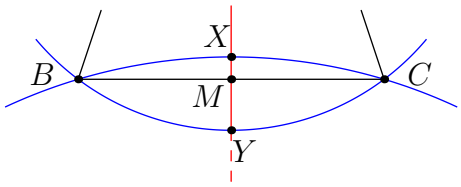
$$\left|\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}\right| = \left|\frac{16(-4)+8(-3)+3(4)}{8}\right|$$
$$= \frac{19}{2} \implies 19+2 = \boxed{21}.$$

AR 5: ABC is an isosceles triangle with base $BC = 6$ and $AB = AC$. Point M is the midpoint of BC such that $AM = 9$. Circle ω_1 is the circumcircle of ABC with radius R, and ω_2 is a circle passing through B and C with radius $2R$ and center on the opposite side of BC from A. Line AM intersects ω_2 at point X and ω_1 at point Y, where X lies between A and Y. The length XY can be expressed as $m-\sqrt{n}$, where m and n are positive integers. Find $m+n$.

Answer: $\boxed{102}$

Solution:





Suppose line AM intersects ω_2 again at $Z \neq X$. Note that because $\triangle ABC$ is isosceles, AY and XZ are diameters of ω_1 and ω_2, respectively, so $XZ = 2 \cdot AY$. Now, by power of a point on M with respect to ω_1, we have $AM \cdot YM = BM \cdot CM$. Since $AM = 9$ and $BM = CM = 3$, we have $YM = 1$. Therefore, $AY = 10$ and $XZ = 20$. Let $t = XM$; note that $ZM = 20 - t$. By power of a point on M with respect to ω_2, we have $XM \cdot ZM = BM \cdot CM$, or $t(20-t) = 9$. Solving, we have $t = 10+\sqrt{91}$ or $t = 10-\sqrt{91}$. Since X lies on segment AM while the center of ω_2 lies on the other side of BC, M lies between X and the center of ω_2. Thus, $t < 10$, so $t = 10-\sqrt{91}$. So,

$$XY = XM + YM = 11-\sqrt{91} \implies 11+91 = \boxed{102}.$$

AR 6: Andy the ant starts at the square labeled 1. On each move, Andy moves to a square that shares a side with his current square, chosen uniformly at random. What is the expected number of moves before Andy is in the square labeled 2?

		2
1		

Answer: $\boxed{18}$

Solution:

8	9	2
5	6	7
1	3	4

Label each cell of the grid as shown, and let e_i denote the expected number of moves Andy makes starting from cell i before reaching cell 2, so $e_2 = 0$ and our answer is e_1. By symmetry, $e_3 = e_5$, $e_4 = e_8$, and $e_7 = e_9$. Since Andy moves to each orthogonally adjacent cell with equal probability, we have

$$\begin{aligned} e_1 &= e_3 + 1 \\ e_3 &= \frac{1}{3}e_1 + \frac{1}{3}e_4 + \frac{1}{3}e_6 + 1 \\ e_4 &= \frac{1}{2}e_3 + \frac{1}{2}e_7 + 1 \\ e_6 &= \frac{1}{2}e_3 + \frac{1}{2}e_7 + 1 \\ e_7 &= \frac{1}{3}e_4 + \frac{1}{3}e_6 + 1. \end{aligned}$$

The third and fourth equations give $e_6 = e_4$. Substituting this and $e_1 = e_3 + 1$ into the second and fifth equations, we have

$$e_3 = e_4 + 2 \quad \text{and} \quad e_7 = \frac{2}{3}e_4 + 1.$$

Substituting these into the third equation, we have

$$e_4 = \frac{1}{2}(e_4 + 2) + \frac{1}{2}\left(\frac{2}{3}e_4 + 1\right) + 1 \implies e_4 = 15,$$

so $e_1 = e_3 + 1 = e_4 + 3 = \boxed{18}$.

AR 7: There is a unique ordered triple of positive reals (a, b, c) satisfying the system of equations

$$\begin{aligned} a^2 + 9 &= (b - 8\sqrt{3})^2 + 4 \\ b^2 + 4 &= (c - 8\sqrt{3})^2 + 49 \\ c^2 + 49 &= (a - 8\sqrt{3})^2 + 9. \end{aligned}$$

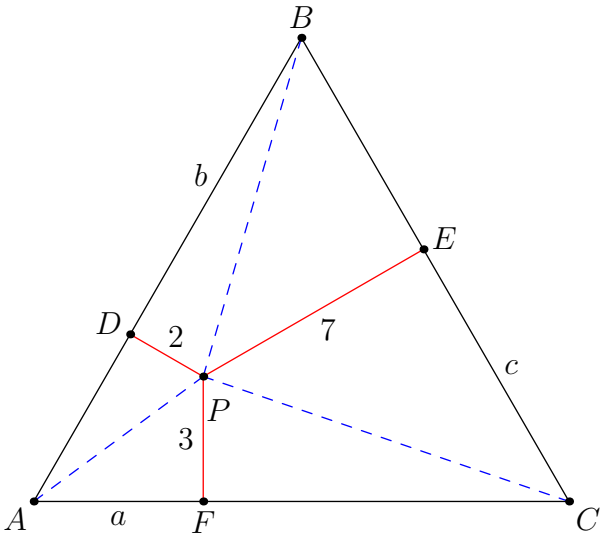
The value of $100a + 10b + c$ can be expressed as $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Answer: $\boxed{294}$

Solution: Rewrite the given system of equations as

$$a^2 + 3^2 = (8\sqrt{3} - b)^2 + 2^2$$
$$b^2 + 2^2 = (8\sqrt{3} - c)^2 + 7^2$$
$$c^2 + 7^2 = (8\sqrt{3} - a)^2 + 3^2.$$

We provide a geometric interpretation for this system. Let $\triangle ABC$ be equilateral, let P be a point in the interior of $\triangle ABC$, and let D, E, and F be the projections of P onto AB, BC, and CA, respectively. Furthermore, suppose that $PD = 2$, $PE = 7$, and $PF = 3$.



We claim that $(a, b, c) = (AF, BD, CE)$. To prove this, we need to verify that the side length s of $\triangle ABC$ is $8\sqrt{3}$. We determine the value of s by computing the area of $\triangle ABC$ in two different ways. On one hand, $[ABC] = \frac{1}{4}s^2\sqrt{3}$. On the other hand, $[ABC] = [APB] + [BPC] + [CPA] = 6s$. Equating these two expressions for $[ABC]$, we find $s = 8\sqrt{3}$, as desired. Now, we use the Law of Cosines on $\triangle FAD$ and $\triangle FPD$ to obtain two expressions for the value of DF^2 that we can set equal. Note that $\angle FAD = 60^\circ$ and $\angle FPD = 120^\circ$. Thus, we have

$$DF^2 = a^2 + (8\sqrt{3} - b)^2 - a(8\sqrt{3} - b) = 19.$$

We also know that $(8\sqrt{3} - b)^2 = a^2 + 5$, and because $b = BD < 8\sqrt{3}$, we have $8\sqrt{3} - b = \sqrt{a^2 + 5}$. Plugging in these expressions for $8\sqrt{3} - b$ and $(8\sqrt{3} - b)^2$ into the above equation and simplifying gives us

$$2a^2 - 14 = a\sqrt{a^2 + 5}.$$

Squaring and rearranging this equation yields the even quartic $3a^4 - 61a^2 + 196 = 0$. Solving and discarding negative solutions, we get $a = 2$ or $a = \frac{7}{\sqrt{3}}$. However, $a = 2$ is extraneous as $2(2)^2 - 14 \neq 2\sqrt{2^2 + 5}$, so $a = \frac{7}{\sqrt{3}}$. Substituting back into the system, we have $(a, b, c) = (\frac{7}{\sqrt{3}}, \frac{16}{\sqrt{3}}, \frac{13}{\sqrt{3}})$. So,

$$100a + 10b + c = \frac{873}{\sqrt{3}} = 291\sqrt{3} \implies 291 + 3 = \boxed{294}.$$

AR 8: We say that a permutation $(a_1, a_2, \ldots, a_{10})$ of the integers 1 through 10 inclusive is *peaked* if there do not exist three integers $1 \leq i < j < k \leq 10$ such that $a_i > a_j$ and $a_j < a_k$. Let $\mathcal{S}$ be the set of all peaked permutations. If $a_p = 9$ and $a_q = 4$, the expected value of $|p - q|$ over all permutations in $\mathcal{S}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{9}$

Solution: Let $a_r = 10$. We claim that a permutation is peaked if and only if the terms before 10 are increasing and the terms after 10 are decreasing. If $a_i > a_j$ for some $i < j < r$, then $a_i > a_j$ and $a_j < a_r$, so the terms before 10 must be increasing. Similarly, the terms after 10 must be decreasing. Conversely, in such a permutation, every term is greater than all the terms on one side of it, so it is peaked. So, each peaked permutation corresponds to choosing, for each of $1, 2, \ldots, 9$, whether it lies before or after 10, and each of these 2^9 choices is equally likely. Now, $|p-q|-1$ is the number of terms strictly between 9 and 4. If 4 and 9 lie on the same side of 10, then a term d lies between them if and only if $4 < d < 9$ and d lies on the same side, so the expected number of such terms is $4 \cdot \frac{1}{2} = 2$. Otherwise, 10 lies between them, and since no term other than 10 is greater than 9, a term $d \neq 10$ lies between them if and only if $4 < d < 9$ and d lies on the same side as 4, so the expected number of such terms is $1 + 4 \cdot \frac{1}{2} = 3$. Each case occurs with probability $\frac{1}{2}$, so

$$\mathbb{E}(|p-q|) = 1 + \frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 3 = \frac{7}{2} \implies 7 + 2 = \boxed{9}.$$

AR 9: For a positive integer n, let $r(n)$ denote the value of the binary number obtained by reading the binary representation of n from right to left. For example, $r(6) = r(110_2) = 011_2 = 3$. If k is the smallest positive integer such that the equation $n + r(n) = 2k$ has at least ten positive integer solutions n, find k.

Answer: $\boxed{768}$

Solution: Let n have L binary digits $b_{L-1}b_{L-2}\ldots b_0$, where $b_{L-1} = 1$. Since $r(n)$ is odd and $n + r(n) = 2k$ is even, n is odd, so $b_0 = 1$. Let $s_i = b_i + b_{L-1-i}$, so that

$$n + r(n) = \sum_{i=0}^{L-1} s_i \cdot 2^i,$$

where $s_i = s_{L-1-i}$ and $s_0 = s_{L-1} = 2$. Note that $2^L + 2 \leq n + r(n) \leq 2^{L+1} - 2$, so all solutions for a fixed k have the same number of digits. We claim that k also determines every s_i. Suppose two such sequences give the same k, let e_i be their differences, and let j be the smallest index with $e_j \neq 0$. Then, $e_j 2^j + e_{j+1}2^{j+1} + \cdots + e_{L-1-j}2^{L-1-j} = 0$, so e_j is even, meaning $e_j = e_{L-1-j} = \pm 2$ (and $j < L - 1 - j$). However, since $|e_i| \leq 2$, we have

$$2 \cdot 2^{L-1-j} + 2 \cdot 2^j > 2\left(2^{j+1} + 2^{j+2} + \cdots + 2^{L-2-j}\right),$$

a contradiction. So, the number of solutions for a fixed k is 2^m, where m is the number of pairs $i < L - 1 - i$ with $s_i = 1$, since each such pair of digits can be either 01 or 10. Since $s_0 = 2$, we have $m \leq \left\lfloor \frac{L-2}{2} \right\rfloor$. For at least ten solutions, we need $m \geq 4$, so $L \geq 10$. If $L \geq 11$, then $k \geq \frac{2^{11}+2}{2} > 768$. If $L = 10$, we need $s_1 = s_2 = s_3 = s_4 = 1$, giving

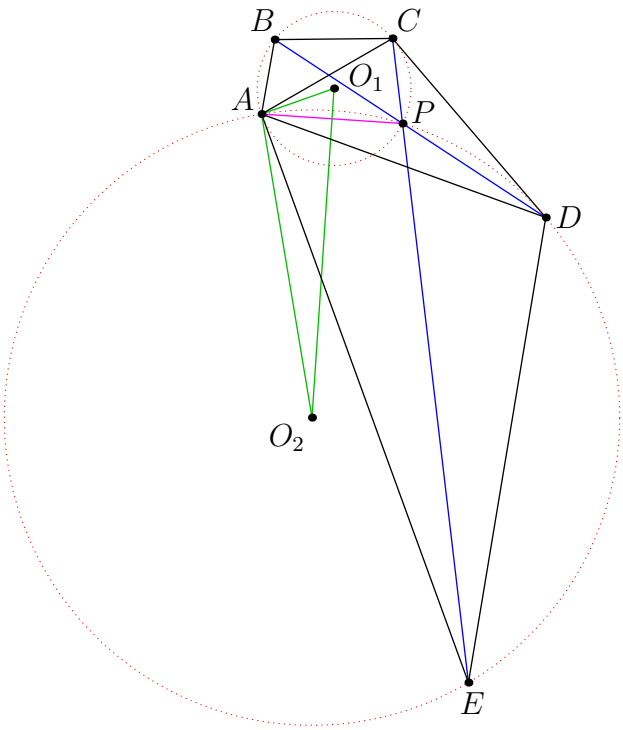
$$2k = 2\left(2^0 + 2^9\right) + \left(2^1 + 2^2 + 2^3 + 2^4\right) + \left(2^5 + 2^6 + 2^7 + 2^8\right) = 1536$$

with $2^4 = 16$ solutions. So, the answer is $k = \boxed{768}$.

AR 10: Let $ABCDE$ be a convex pentagon with $\angle BAC = \angle CAD = \angle DAE$ and $\angle ABC = \angle ACD = \angle ADE$. Let BD and CE meet at P. Given that $BC = 6$, $\sin \angle BAC = \frac{3}{5}$, and $\frac{AC}{AB} = 5$, the length of AP can be expressed as $\frac{m}{\sqrt{n}}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Answer: $\boxed{57}$

Solution:



From the given angle equalities and AA similarity, we have $\triangle ABC \sim \triangle ACD \sim \triangle ADE$. Let $\mathcal{T}$ be the transformation that sends $\triangle ABC$ to $\triangle ACD$. Since $\frac{AC}{AB} = 5$, $\mathcal{T}$ is a rotation by $\angle BAC$ about A followed by a dilation with scale factor 5 centered at A. We claim that P is the second intersection point of (ABC) and (ADE). Note that $\mathcal{T}$ sends BD to CE, so $\angle BPC = \angle BAC$. This implies that P lies on (ABC). Since $\angle DPE = \angle BPC = \angle BAC = \angle DAE$, P also lies on (ADE). Next, let O_1 and O_2 be the centers of (ABC) and (ADE), respectively. We seek the value of $\cos(\angle O_1AO_2)$. Note that $\mathcal{T}^2$ sends $\triangle ABC$ to $\triangle ADE$, so $\mathcal{T}^2$ sends O_1 to O_2. (Here, $\mathcal{T}^2$ denotes $\mathcal{T}$ applied twice.) Therefore, $\angle O_1AO_2 = 2\angle BAC$. Then,

$$\begin{aligned}\cos(\angle O_1AO_2) &= \cos(2\angle BAC)\\ &= 1 - 2\sin^2(\angle BAC)\\ &= \frac{7}{25}.\end{aligned}$$

By the extended Law of Sines, the radius of (ABC) is $\frac{BC}{2\sin(\angle BAC)} = 5$, which means that

the radius of (ADE) is $25 \cdot 5 = 125$. By the Law of Cosines on $\triangle O_1AO_2$, we have

$$\begin{aligned}(O_1O_2)^2 &= (AO_1)^2 + (AO_2)^2 - 2(AO_1)(AO_2)\cos(\angle O_1AO_2) \\ &= 5^2 + 125^2 - 2(5)(125)\left(\frac{7}{25}\right) \\ &= 15300,\end{aligned}$$

so $O_1O_2 = 30\sqrt{17}$. Finally, to find AP, we compute $[O_1AO_2]$. We have $\sin(\angle O_1AO_2) = \sqrt{1-\cos^2(\angle O_1AO_2)} = \frac{24}{25}$, so from the sine area formula,

$$[O_1AO_2] = \frac{1}{2}(5)(125)\left(\frac{24}{25}\right) = 300.$$

Now, since O_1O_2 is the perpendicular bisector of AP, the length of the altitude from A to O_1O_2 is $\frac{AP}{2}$. Thus, $[O_1AO_2] = \frac{1}{2}(30\sqrt{17})(\frac{AP}{2}) = 300$. Solving, we have

$$AP = \frac{40}{\sqrt{17}} \implies 40 + 17 = \boxed{57}.$$